Status	Finished
Started	Sunday, 1 November 2026, 7:41 PM
Completed	Sunday, 1 November 2026, 7:45 PM
Duration	4 mins 49 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n/2
c. n-1
d. n ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)
b. Hash search
c. Linear search ✓
d. Binary search

Question **3**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Sequential search
c. Hash search
d. Binary search ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. The last element is returned
c. The search is considered unsuccessful ✓
d. An error is raised

Question **5**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues from the beginning
c. The search continues in the left or right sublist ✓
d. The list is sorted

Question **6**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓
b. At the beginning of the list
c. In the middle of the list
d. In the left sublist

Question **7**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Bubble ✓

c. Insertion

d. Selection

Question **8**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Interpolation search

b. Hash search

c. Linear search ✓

d. Binary search

Question **9**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list is sorted

b. When the element is found in the list ✓

c. When the search algorithm finishes

d. When the list contains duplicate elements

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. 1
c. n/2
d. n ✓

Question **11**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list
b. It can be applied only if the list is sorted ✓
c. It works on unsorted lists
d. It compares elements sequentially

Question **12**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Mining
c. Finding
d. Searching ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list is unsorted
c. When the list is very small
d. When the list contains duplicate elements

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It does not work with negative numbers
c. It can only be applied to large lists
d. It requires the list to be sorted ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Checking each element sequentially ✓
b. Skipping every second element
c. Sorting the list before searching
d. Dividing the list in half repeatedly